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Three-Dimensional Ultrasonic Reconstruction and Landing-Point Prediction of a Thrown Projectile

*Trilateration, Kalman filtering and parabolic extrapolation from three
time-multiplexed HC-SR04 range sensors, with an empirical throw-aligned drag
correction*

Final Project Report

Instructor: Prof. Zappa

Target: size-7 basketball | 31 ground-truthed throws | 242 automated tests passing

9 July 2026

Abstract

This project develops and validates a low-cost measurement system that reconstructs the three-dimensional trajectory of a projectile thrown over a small sensor array and predicts its landing point on the floor. Three HC-SR04 ultrasonic transducers are placed at the vertices of an equilateral triangle of side $s = 1.000$ m, each tilted 55° above the horizontal and aimed radially inward. The sensors are fired in a rigid 18 ms time-multiplexed schedule by an Arduino Nano Every, which streams raw echo times to a host computer. In post-processing, each triplet of ranges is converted to distance from the session temperature, corrected for the ball radius, per-sensor electronic offset and inter-sensor sampling delay, and solved for the ball position by trilateration. A seven-state Kalman filter with a gravity-encoding augmented state smooths the track; a time-parameterised parabola is fitted and extrapolated to the contact height $y = r_{\text{ball}}$, yielding the landing point in polar coordinates (r, θ) with a per-throw uncertainty ellipse. A separate offline programme calibrates a throw-aligned empirical drag correction across a campaign of throws.

Bench characterisation revealed that the originally-proposed tennis ball is acoustically invisible at 40 kHz (persistent time-outs), so — with the instructor’s approval — the target was changed to a basketball, whose hard, quasi-smooth shell returns an echo roughly an order of magnitude stronger. Over a campaign of **31** ground-truthed throws spanning the full heading range, the raw landing error was **93 mm** mean (median 95 mm, std 43 mm). The trajectory carries a statistically significant along-track bias of **68 mm** (95% CI [46, 89] mm), consistent with aerodynamic drag and the parabola-extrapolation geometry; the empirical correction reduces the mean error on held-out throws from **90 mm to 56 mm** ($p \approx 1 \times 10^{-4}$). The complete software pipeline is validated end-to-end against a physics-based simulator and ships with 242 passing automated tests.

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1. Introduction and objectives

The goal of this project is to **measure a physical trajectory and predict a future event** from it — where a thrown ball will strike the floor — using only inexpensive ultrasonic range sensors. The problem is a compact illustration of the full measurement chain taught in the course: transducer physics, timing and synchronisation, unit conversion under environmental influence quantities, geometric inversion, state estimation, propagation of uncertainty, and the separation of random error from systematic error through calibration.

The concrete deliverables were: (i) a three-sensor array whose geometry is surveyed and fixed; (ii) firmware that ranges the ball repeatedly during flight without timing drift; (iii) a processing pipeline that reconstructs the 3-D track and extrapolates the landing point in polar coordinates (r , θ) referred to the array centroid; (iv) a per-throw uncertainty estimate; and (v) an empirical, statistically-gated drag correction calibrated over a campaign of throws with measured ground truth.

1.1 Measurement principle

An ultrasonic pulse travels to the ball and back; the round-trip time t_{echo} gives the range once the speed of sound is known. Three simultaneous-enough ranges to known sensor positions locate the ball in space (trilateration). Sampling the ball repeatedly as it flies produces a sequence of 3-D points; because the only appreciable force in flight is gravity, the vertical coordinate is a parabola in time and the horizontal coordinates are very nearly linear, so the track can be fitted and extrapolated to the instant the ball reaches contact height. The predicted (x , z) is then expressed as range and bearing from the array centre.

1.2 Scope and headline result

The system was built, bench-characterised on real hardware, and exercised over a campaign of 31 ground-truthed live throws. All physical constants and thresholds live in a single configuration file; nothing physical is hard-coded in firmware or in the analysis. The end-to-end software is validated against a physics-based simulator through an automated acceptance gate. The headline results are summarised in Table 1.

Quantity	Value
Array	Equilateral triangle, side $s = 1.000$ m; sensors tilted 55° inward
Controller / sensors	Arduino Nano Every + $3 \times$ HC-SR04 (40 kHz)
Target	Size-7 basketball, $r = 119.4$ mm, $m = 0.620$ kg (approved substitution)
Ground-truthed throws	31 (headings spanning the full $\pm 180^\circ$)
Triplets used per throw	4–6 (mean 4.6)
Raw landing error (2-D)	mean 93 mm, median 95 mm, std 43 mm
Along-track bias (drag)	68 mm, 95% CI [46, 89] mm (significant)
Error after correction	$90 \rightarrow 56$ mm on held-out throws ($p \approx 1e-4$)
Per-throw uncertainty	mean $\sigma_r \approx 44$ mm, $\sigma_\theta \approx 2.9^\circ$ (1 dof, indicative)
Software	242 automated tests passing; simulator-validated acceptance gate

Table 1. System and campaign at a glance.

2. System architecture and coordinate frame

The system separates cleanly into a real-time acquisition half (firmware + serial capture) and an offline analysis half (a chain of pure, independently-tested computational modules driven by a single orchestrator). Figure 1 defines the coordinate frame and Figure 2 shows the processing chain.

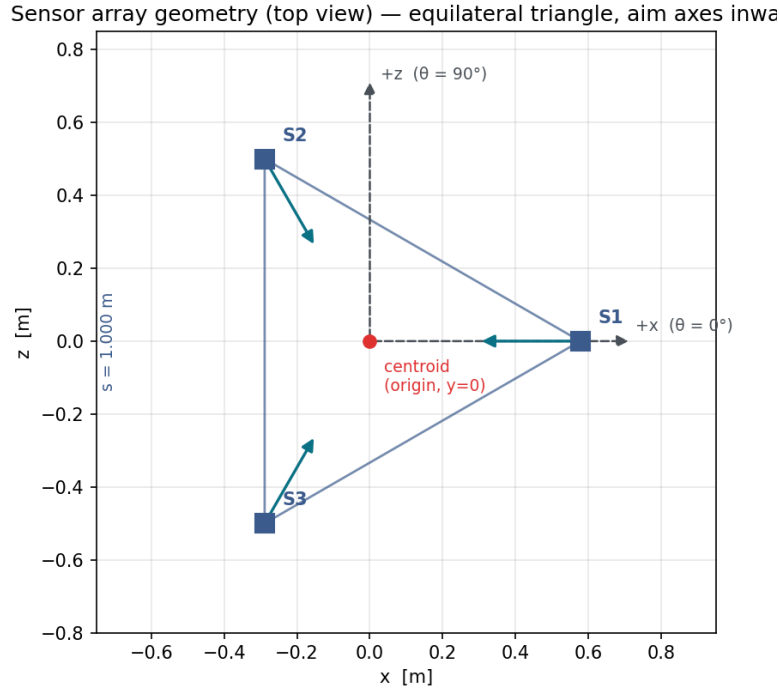


Figure 1. Array geometry and coordinate frame (top view). The origin is the array centroid, on the floor ($y = 0$). $+x$ points toward sensor S1 and is the $\theta = 0$ reference; $+z$ completes a right-handed frame with $+y$ up. Cyan arrows are the inward sensor aim axes; each sensor is also tilted 55° out of the page. Landing is reported as (r, θ) from the centroid.

Coordinate conventions. All mathematics is performed in Cartesian coordinates; polar output ($r = \sqrt{x^2 + z^2}$, $\theta = \text{atan2}(z, x)$ in degrees) is produced only at the final reporting step, and ground-truth tape measurements are converted to Cartesian on input. Fitting or averaging is never done in polar coordinates, where the wrap at $\pm 180^\circ$ and the angle-range coupling would corrupt statistics. The acoustic centres of the three transducers sit at a common height S_{height} above the floor; trilateration is solved in the sensor plane and the floor frame is recovered by $y_{\text{floor}} = y_{\text{plane}} + S_{\text{height}}$. Equal sensor height is an explicit, surveyed assumption (agreement required within ± 3 mm).

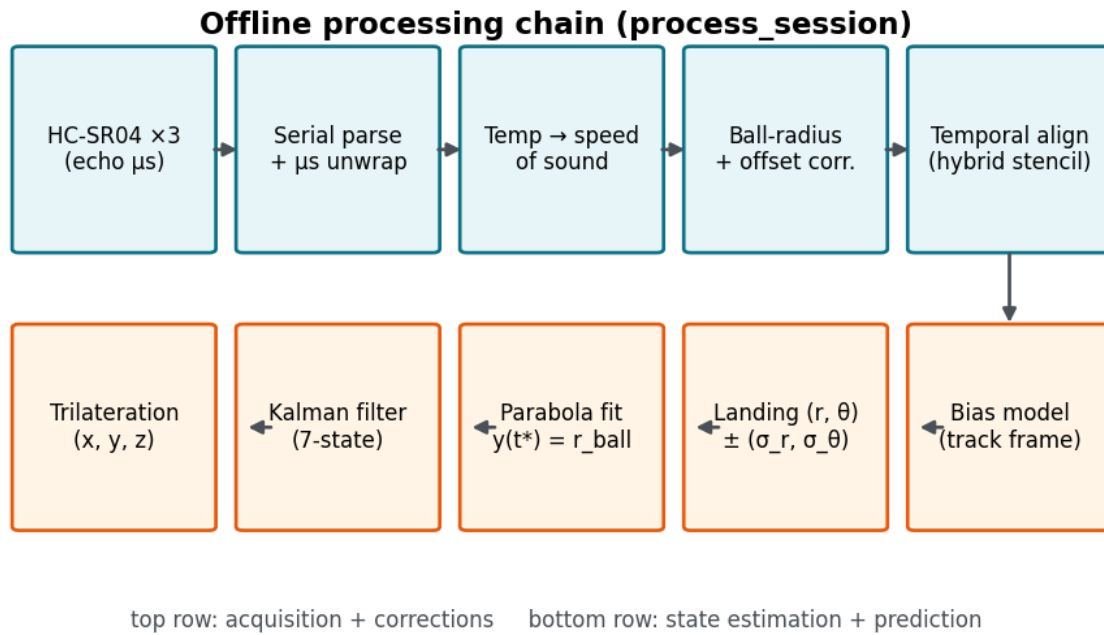


Figure 2. Offline processing chain. Raw echo times enter at top left and flow through unit conversion and the three geometric corrections, trilateration, Kalman smoothing, parabolic landing extrapolation and the optional empirical bias correction. Each computational block is a pure, separately unit-tested module.

3. Hardware and firmware

3.1 Sensors, controller and wiring

Each HC-SR04 emits a 40 kHz burst on a 10 μ s trigger and returns an echo pulse whose width equals the round-trip time. The three sensors share the 5 V rail and a common ground; trigger and echo lines use dedicated digital pins (S1→D2/D3, S2→D4/D5, S3→D6/D7). Leads are bundled and labelled per sensor and tug-tested, because a swapped sensor identity silently corrupts trilateration. The controller is an Arduino Nano Every; the firmware uses only `digitalWrite`, `pulseIn` and `micros()` — no interrupts or timer libraries — so its timing behaviour is transparent and reproducible.

3.2 Timing discipline

The single most important firmware property is a **rigid, drift-free schedule**. Each sensor owns an exact 18 ms slot governed by an absolute scheduler (`next_slot += 18000  $\mu$ s`), so one full triplet takes 54 ms and the per-sensor sampling instants are deterministic. The echo time-out is 12.5 ms, which covers the full 2 m usable range across the lab temperature band while leaving roughly 5.5 ms of idle margin per slot for formatting and transmission. A time-out returns a literal 0 sentinel. The firmware streams raw integers only — `sensor_id`, `echo_us`, `t_trig_us` — and never computes distance: that is the host’s job, because distance depends on the measured temperature (Section 4).

The rigid schedule is also a deliberate defence against a **ceiling wrap-around ghost**: a static reflector produces a static aliased echo that the background calibration captures and the gate subtracts. A jittered schedule would smear that ghost into untrackable noise, so the schedule is kept fixed by design. Firmware was developed as five incremental sketches, each with its own bench pass criterion, culminating in a PC-side timing verification that asserts a 54.000 ms period to ± 50 μ s, correct inter-sensor phase, strictly monotonic time stamps after roll-over unwrap, and zero parse errors.

Parameter	Value	Rationale
Slot per sensor	18 ms	Absolute scheduler; no cumulative drift
Triplet period	54 ms	≈ 5–6 samples over a 250–330 ms flight
Echo time-out	12.5 ms	Covers 2 m round-trip at 5–20 °C; widens idle margin
Max gated range	2.0 m	Software gate; echoes beyond are discarded
Baud	115200	≈ 1.4 kB/s « link capacity; no buffering risk
Sample instant	$t_{\text{trig}} + \text{echo}/2$	Mid-echo: ball is interrogated when the pulse reaches it

Table 2. Firmware timing and link parameters.

4. From echo time to ball position

4.1 Speed of sound and range

The host prompts for the ambient temperature T [°C] and computes the speed of sound and range as

$$v = 331.4 + 0.606 \cdot T \text{ [m/s]}, \quad d = v \cdot t_{\text{echo}} \cdot 10^{-6} / 2 \text{ [m]}.$$

The residual humidity and pressure dependence is at the 0.1–0.3% level (millimetres at these ranges) and is accepted rather than self-calibrated. Distances exist only on the host; the firmware never sees metres.

4.2 Three geometric corrections

Three small but load-bearing corrections are applied to every valid reading before trilateration:

- **Ball-radius (surface→centre).** The echo returns from the nearest surface point, which lies on the sensor–centre line, so the true centre range is $d_{\text{centre}} = d_{\text{raw}} + r_{\text{ball}}$. For the basketball this adds 119.4 mm — about 12% of a 1 m range — and is therefore the single largest correction. It is exact for a sphere.
- **Per-sensor electronic offset.** A constant per-sensor bias (transducer/timing) measured on a flat board is subtracted at the acquisition layer (Section 6). This is a measurement-layer correction, kept deliberately separate from the empirical drag model.
- **Mid-echo / temporal alignment.** The ball is interrogated when the pulse reaches it, at $t_{\text{sample}} = t_{\text{trig}} + t_{\text{echo}}/2$, not at the trigger instant. Because the three sensors fire 18 ms apart, each triplet’s ranges are sampled at slightly different times; a hybrid finite-difference stencil estimates each sensor’s radial range-rate and extrapolates all three ranges back to a common instant, cancelling the leading motion-induced error term.

The temporal stencil is **forward** at the first triplet, **central** in the interior and **backward** at the last, so all N triplets are retained — important when a throw yields only 4–6 of them. The central interior differences cancel the leading curvature bias; the residual along-track systematic is about –12 to –15 mm and is carried explicitly in the error budget (Section 9) rather than hidden.

5. Trilateration, filtering and landing prediction

5.1 Trilateration

With three sensors at horizontal positions (X_i, Z_i) at equal height, subtracting the S1 sphere equation from the S2 and S3 spheres cancels the quadratic and the height terms, leaving a linear 2×2 system for the horizontal position:

$$\mathbf{A} \cdot [\mathbf{x}; \mathbf{z}] = \mathbf{b}, \quad \mathbf{A} = 2 \cdot \begin{bmatrix} X_2 - X_1 & Z_2 - Z_1 \\ X_3 - X_1 & Z_3 - Z_1 \end{bmatrix}; \quad \mathbf{b} = \begin{bmatrix} d_1^2 - d_2^2 + (X_2^2 - X_1^2) + (Z_2^2 - Z_1^2) \\ d_1^2 - d_3^2 + (X_3^2 - X_1^2) + (Z_3^2 - Z_1^2) \end{bmatrix}.$$

The matrix **A** depends only on the fixed sensor positions, so it is constant and well-conditioned for an equilateral array; the trajectory enters only through **b**. The height above the sensor plane is then recovered from all three spheres,

$$y_k = \sqrt{(d_k^2 - (x - X_k)^2 - (z - Z_k)^2)}, \quad k = 1, 2, 3,$$

averaged over the three sensors, with a strict guard: if any radicand is negative the triplet is physically impossible and is flagged and discarded (never clamped). The solution is anisotropic in noise — a geometric dilution of precision (GDOP) of about 2.3 for the equilateral array — but never singular. Symmetric-corridor throws (along a symmetry axis) weaken depth resolution, so the campaign deliberately varied heading off the axes.

5.2 Kalman filter

A seven-state Kalman filter $[x, y, z, v_x, v_y, v_z, 1]$ smooths the reconstructed points. The seventh state is a constant that lets the transition matrix inject gravity exactly; the transition matrix is **rebuilt at every step from the measured Δt** between consecutive used triplets, so gate rejections that create 108–162 ms gaps are handled correctly rather than assuming a fixed period:

$$\mathbf{F}(\Delta t) = \begin{bmatrix} 1 & 0 & 0 & \Delta t & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t & 0 & -\frac{1}{2}g \cdot \Delta t^2 \\ 0 & 0 & 1 & 0 & 0 & \Delta t & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & -g \cdot \Delta t \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

measurement: $\mathbf{H} = [\mathbf{I}_3 \mid \mathbf{0}]$
heteroscedastic $\mathbf{R}$ per update:
 $\mathbf{R} = \text{diag}(\sigma_{\text{pos}}^2, \sigma_y^2, \sigma_{\text{pos}}^2),$
 $\sigma_y = (d^- / y) \cdot \sigma_{\text{pos}}$
 $g = 9.81 \text{ m/s}^2$; 7th state $\equiv 1$ (gravity only)

The measurement covariance is per-triplet: the vertical component is inflated as the ball nears the overlap floor, because the vertical dilution grows there. The gravity state is never used to absorb drag — drag is handled separately and empirically (Section 7) — keeping the filter’s physical model honest.

5.3 Landing prediction and per-throw uncertainty

The smoothed track is fitted in time: $x(t)$ and $z(t)$ as degree-1 polynomials and $y(t)$ as degree-2 (gravity makes it genuinely quadratic). The landing time t^* is the descending-branch root of $y(t^*) = r_{\text{ball}}$ (contact of the sphere bottom with the floor), and the landing point is $(x(t^*), z(t^*))$. Time-parameterisation is chosen over x -parameterisation because the $\pm 60^\circ$ heading spread shrinks the x -span and would ill-condition the fit, whereas time advances monotonically. A concave-down guard rejects a flat or rising fit that would extrapolate to an absurd root.

The coefficient covariance is formed by hand as $(\text{RSS}/\text{dof}) \cdot (\mathbf{V}^T \mathbf{V})^{-1}$ and propagated through t^* by the delta method to a 2×2 landing covariance, from which $(\sigma_r, \sigma_\theta)$ follow. With only 4–6 points and three parameters this rests on 1–3 degrees of freedom, so the ellipse is reported with an explicit small-sample caveat: it is indicative, not a calibrated confidence region. Across the campaign the mean per-throw values are $\sigma_r \approx 44 \text{ mm}$ and $\sigma_\theta \approx 2.9^\circ$.

6. Static characterisation and target substitution

Before any throw, the sensors were characterised against a stationary target at tape-measured distances, using the tape-to-centre convention (tape to the nearest surface point, plus the ball radius). This step proves the electronics, measures the noise that feeds the Kalman covariance, extracts the per-sensor electronic offsets, and — critically — tests whether the intended target is even detectable.

6.1 The tennis ball fails; the basketball is adopted

A flat board was detected cleanly by all three sensors at 1.0 m. The **tennis ball, however, produced persistent time-outs** at 1.0 m on-axis: its felt surface is a fibrous, porous absorber at 40 kHz (wavelength ≈ 8.6 mm), a poor specular reflector. This was the project's top-ranked risk, and it materialised. Following the pre-agreed escalation protocol the substitution was **escalated to and approved by Prof. Zappa**, and the target was changed to a size-7 basketball. Acoustically the basketball is hard (pressurised rubber/leather) and its pebble texture (≈ 1 mm) is well below the wavelength, so the surface is quasi-smooth; the backscatter cross-section scales as πr^2 , giving roughly a thirteen-fold ($\approx +11$ dB) stronger return and a wider effective detection lobe. The ball radius (119.4 mm) and mass (0.620 kg) were measured; both the surface→centre correction and the landing contact condition $y = r_{\text{ball}}$ are exact for any sphere, so no algorithm changed.

6.2 Repeatability, offsets and the narrow detection cone

Table 3 collects the static results. Two findings drive the rest of the report. First, the range repeatability is a few millimetres, and the conservative value $\sigma_{\text{pos}} = 10$ mm was adopted for the Kalman range noise (Figure 3). Second, at 1.0 m the ball was seen **only by the sensor on whose axis it sat**; the other two timed out completely. The effective detection cone is therefore narrow, which is the physical reason a flight yields only about five usable triplets — a constraint designed into the simulator and reflected in the campaign statistics.

Target / range	Sensor	σ [mm]	time-out	offset [mm]
Board 1.0 m	S1	1.2	0 %	−25.5
Board 1.0 m	S2	1.6	0 %	−22.5
Board 1.0 m	S3	5.5	0 %	−12.9
Ball 1.0 m (on-axis)	S1	7.7	0 %	+11.6
Ball 1.0 m (on-axis)	S2	12.9	0 %	+30.8
Ball 1.0 m (on-axis)	S3	9.2	0 %	+48.5
Ball 1.3 m (on-axis)	S1 / S2 / S3	3.2 / 3.7 / 15.6	0 %	+31 / +7 / +16
Ball 1.7 m (on-axis)	S1 / S2 / S3	4.7 / 3.1 / 9.0	11 / 0 / 0 %	+20 / +24 / +21

Table 3. Static characterisation ($n = 200$ per sensor per position). Board offsets (measured against all three sensors simultaneously) are adopted as the electronic offsets. Off-axis sensors timed out 100% at every ball position and are omitted. The flat-board S3 offset (−12.9 mm) and its ball-target offsets (+16...+48 mm) disagree in sign — a documented alignment caveat for S3.

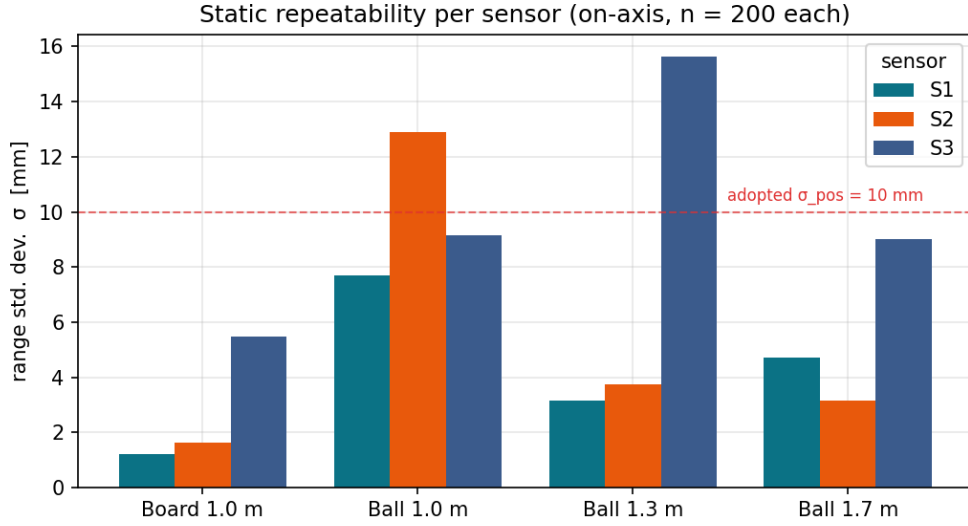


Figure 3. Per-sensor range repeatability (standard deviation of 200 readings) for the flat board at 1.0 m and the basketball at 1.0/1.3/1.7 m, on-axis. The dashed line marks the conservative $\sigma_{\text{pos}} = 10$ mm adopted for the Kalman filter; S3 at 1.3 m is the noisiest point and is the sensor flagged for the offset-sign caveat.

The board-derived offsets (S1 -25.5 , S2 -22.5 , S3 -12.9 mm; a negative offset means the sensor reads short, so the correction adds distance) are applied by default in the live pipeline; a command-line switch disables them for diagnosis. Because a constant per-sensor shift cancels in the background-subtraction band rule, the offset does not perturb the detection gate.

7. Empirical drag correction (throw-aligned)

The reconstruction assumes drag-free, gravity-only flight, but a real throw decelerates and falls short along its direction of travel. Because throws are spread across all headings, a fixed-axis average of the residuals would cancel this real effect. The correction is therefore estimated in a **throw-aligned frame**: for each calibration throw the prediction-truth residual (r_x, r_z) is rotated by the throw heading φ into along-track and cross-track components,

$$r_{\text{along}} = r_x \cos\varphi + r_z \sin\varphi, \quad r_{\text{cross}} = -r_x \sin\varphi + r_z \cos\varphi.$$

A drag overshoot appears as a consistent positive mean in r_{along} with $r_{\text{cross}} \approx 0$. A correction is applied only if its 95% confidence interval — built from Student's t , never a hard-coded 1.96 — excludes zero (a significance gate). Figure 4 shows the residuals in the throw-aligned frame; the along-track cloud is clearly offset in the positive direction while the cross-track component is centred on zero, exactly the drag signature.

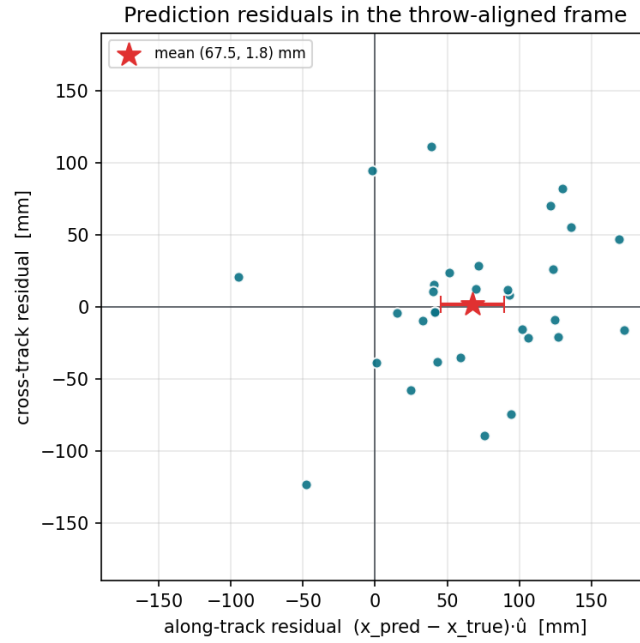


Figure 4. Prediction residuals rotated into the throw-aligned frame. Each point is one throw; the star is the campaign mean with its 95% confidence interval on the along-track axis. The systematic positive along-track offset (drag + extrapolation geometry) is resolved; the cross-track mean is consistent with zero.

The model is validated on a strict disjoint calibrate/validate split before being refit on all throws for use. Table 4 reports the fit; Figure 5 compares mean error before and after correction, both in-sample and on the held-out half. The along-track offset is **67.5 mm** with a 95% CI of [45.7, 89.4] mm, comfortably excluding zero; the cross-track offset is not significant. On the held-out throws the mean 2-D error falls from 90.2 mm to 56.1 mm (one-sided $p \approx 6e-05$), confirming that the correction generalises rather than merely fitting noise.

Quantity	Along-track	Cross-track
Mean residual	67.5 mm	1.8 mm
Std (throw-to-throw)	59.7 mm	51.3 mm
Std. error of mean	10.7 mm	9.2 mm
95% CI	[45.7, 89.4] mm	[-17.0, 20.7] mm
Excludes zero?	YES (significant)	no

Table 4. Throw-aligned bias model over $N = 31$ throws (scalar mode; calibrate/validate split then refit).

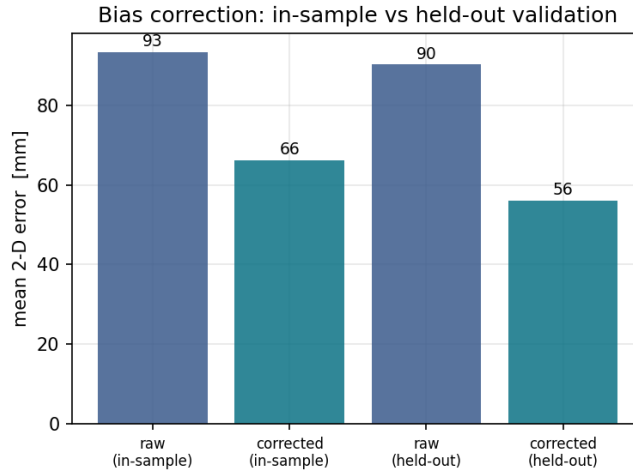


Figure 5. Mean 2-D landing error before and after the empirical correction, shown both in-sample (all throws, descriptive) and on the held-out validation half (unbiased generalisation estimate). The correction removes roughly a third of the mean error.

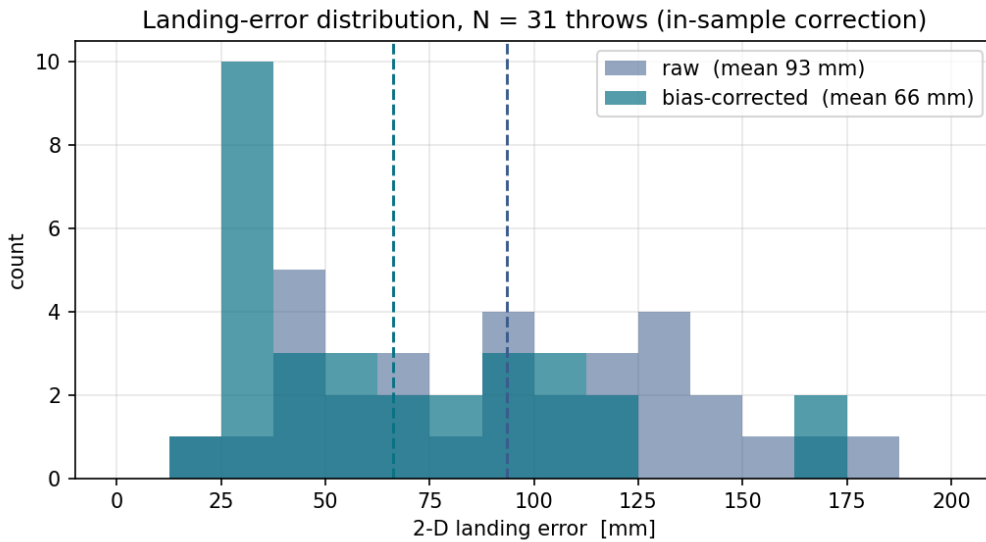


Figure 6. Distribution of the 2-D landing error across the campaign, raw versus bias-corrected (in-sample). The correction shifts the whole distribution toward the origin; the residual spread is the random floor set by the few-triplet fit geometry and range noise.

8. Campaign results

The campaign comprised 31 ground-truthed throws (34 session directories on disk, of which three are re-processed duplicates of the same throws). Headings were spread across the full $\pm 180^\circ$ to exercise the direction-dependent geometry and to give the throw-aligned bias model the heading diversity it needs. Ground truth was measured by first-contact marking and multilateration from three tape distances (the centroid and the two sensors nearest the predicted landing), which gives a unique solution with no mirror ambiguity and a stored least-squares residual as an integrity check. The recorded tape uncertainty of 5 mm propagates to a ground-truth position uncertainty of about 4 mm.

The raw landing error is **93 mm mean** (median 95 mm, std 43 mm, range 16–175 mm). Throws yielded 4–6 usable triplets (mean 4.6), confirming the narrow-cone prediction from the static study. Figures 7–10 present the campaign diagnostics generated by the analysis tool.

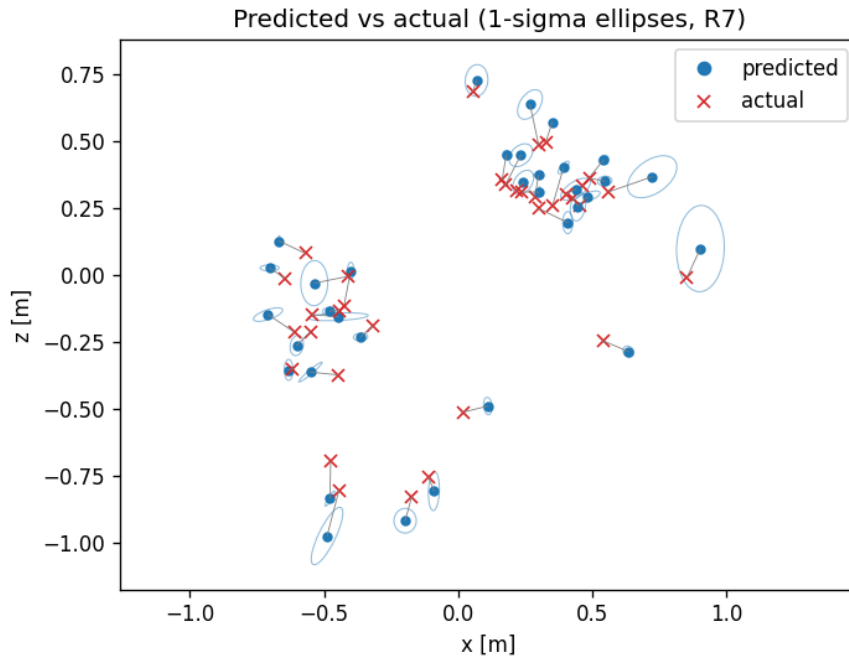


Figure 7. Predicted versus ground-truth landing points across the campaign, with per-throw uncertainty ellipses. Points cluster along the identity line; the systematic radial shortfall visible in several throws is the drag bias removed by the correction of Section 7.

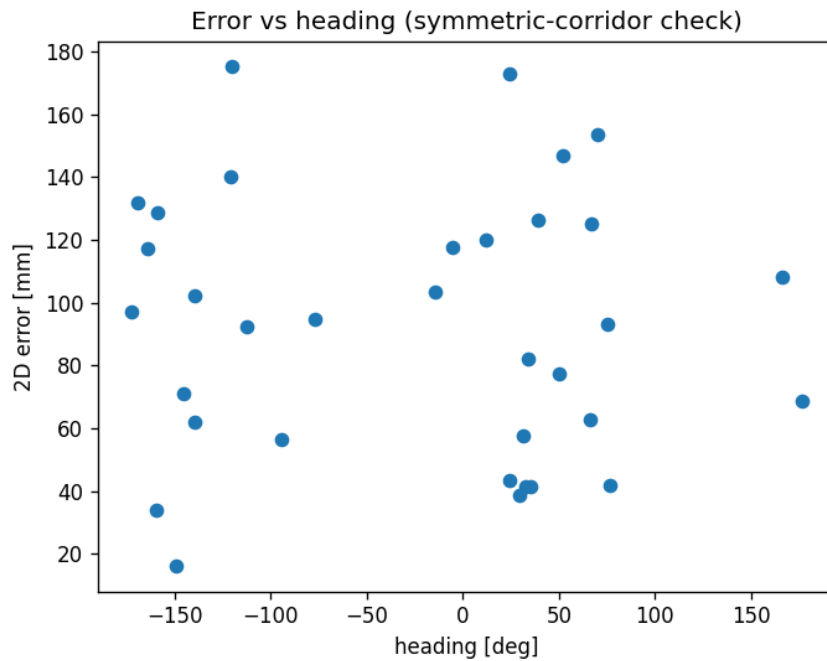


Figure 8. 2-D landing error versus throw heading. The spread confirms the deliberate heading diversity of the campaign and lets the symmetric-corridor (depth-weakening) directions be checked; no single heading dominates the error.

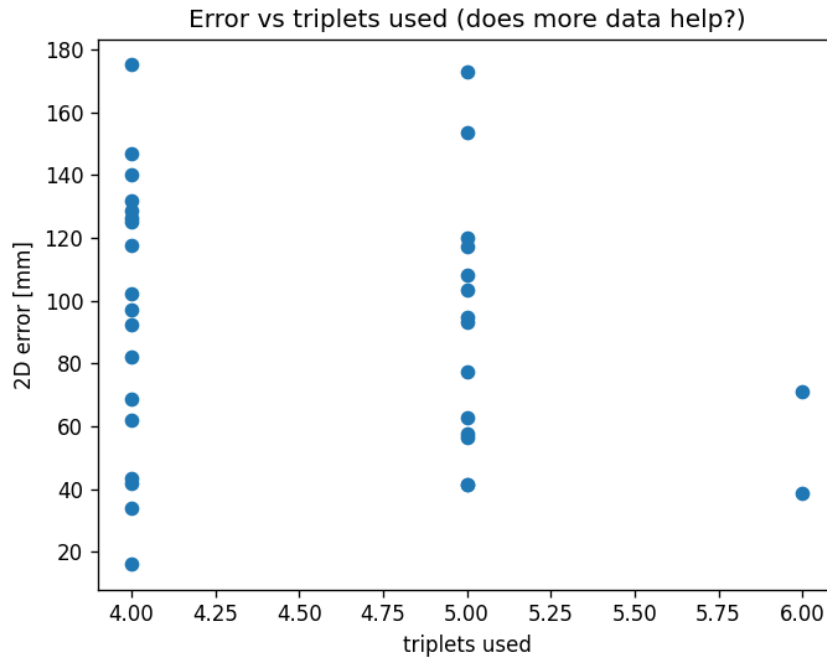


Figure 9. Landing error versus number of usable triplets. Fewer triplets give a less-constrained parabola fit and larger scatter, motivating loftier/slower throws to raise the sample count.

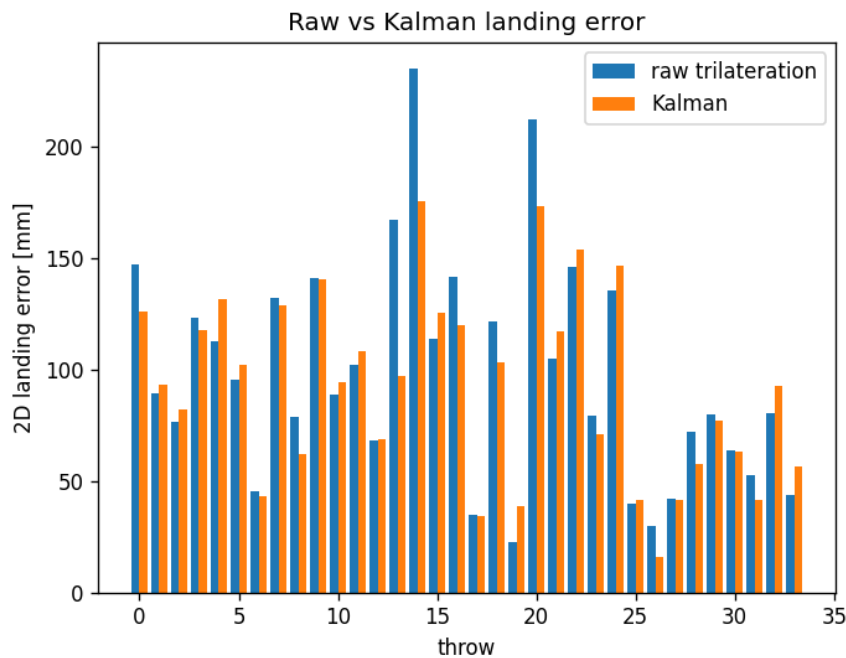


Figure 10. Raw-trilateration versus Kalman-filtered track error. The filter reduces per-point scatter before the parabola fit, most visibly on throws with a rejected mid-flight triplet.

9. Uncertainty and error budget

The measured raw error of about 93 mm decomposes into a correctable systematic part and a random floor. Table 5 gives the budget. The dominant systematic is aerodynamic drag together with the parabola-extrapolation geometry (≈ 68 mm along-track), which the empirical model removes; the random floor (std ≈ 43 mm) is set mainly by the few-triplet fit geometry acting on millimetre-level range noise amplified by the GDOP.

Contribution	Type	Magnitude	Treatment
Aerodynamic drag + extrapolation	systematic	≈ 68 mm along-track	empirical throw-aligned correction (Sec. 7)
Temporal-correction curvature	systematic	≈ -12 to -15 mm	hybrid stencil; residual carried, not hidden
Range repeatability $\times$ GDOP	random	$\sigma_{\text{pos}} 10 \text{ mm} \times \approx 2.3$	Kalman smoothing; averaged over triplets
Few-triplet fit (1-3 dof)	random	per-throw $\sigma_r \approx 44$ mm	reported as uncertainty ellipse; loftier throws
Ball-radius correction	systematic	exact (± 0.3 mm on r)	added before trilateration; negligible residual
Speed of sound (humidity/press.)	systematic	0.1-0.3 % ($\approx 1-3$ mm)	accepted; formula from typed temperature
Sensor-height assumption	systematic	≈ 4 mm if 45 mm omitted	surveyed equal within ± 3 mm; corrected
Ground-truth (tape)	random	≈ 4 mm (5 mm tape)	bounds validation; in the error budget

Table 5. Landing-error budget. Systematic terms are either corrected or bounded; the random floor dominates the residual after drag correction.

After correction the held-out mean error is 56 mm, consistent with this random floor. The per-throw uncertainty ellipses have coverage broadly consistent with their nominal level given the small-sample caveat. Two structural simplifications are stated honestly as model assumptions: equal sensor height (surveyed), and the neglect of the x-y-z cross-correlation in the Kalman measurement covariance.

10. Software verification

Every processing layer is validated against synthetic data before anything is built on top of it. A physics-based simulator generates throws sampled at the physically correct instants (solving the implicit range-timing by fixed-point iteration), applies a beam-cone visibility model so synthetic triplet counts match reality, corrupts the ranges with realistic noise and time-outs, and writes sessions in the identical on-disk format as hardware. The end-to-end acceptance test runs drag-on and drag-off synthetic campaigns through the very same code path as live processing and asserts, among other things, that disabling the ball-radius correction makes the pipeline fail by the expected ≈ 160 mm — a negative control that proves the correction is load-bearing. The full suite is **242 automated tests, all passing**.

11. Limitations, risks and future work

- **Few triplets per throw.** The narrow effective cone yields only 4-6 samples per flight, so the parabola rests on 1-3 degrees of freedom and the per-throw ellipse is only indicative. Loftier, slower throws and a wider array would raise the sample count.
- **Pod-strike hazard.** A 0.6 kg ball landing inside a 1 m triangle can strike a sensor pod; a moved pod invalidates the surveyed vertex coordinates and must trigger a re-survey.
- **Larger surface→centre correction.** The 119 mm radius correction now dominates the range, so any radius mis-measurement propagates one-to-one; the circumference was taped to ± 2 mm.
- **Ceiling wrap-around ghost.** Defended by the rigid schedule and background band rule; verified at survey time from an entered ceiling height.

- **Statistical power.** The realistic drag bias is comparable to the throw-to-throw scatter, so a campaign of 15–30 throws is needed for the confidence interval to exclude zero; this was achieved.
- **Future work.** A four-sensor array would add redundancy and a genuine consistency residual; a tightly-coupled extended Kalman filter on raw ranges is a known-better architecture; and an optical cross-check module is specified for future field work.

12. Conclusions

A complete ultrasonic measurement chain was designed, built on real hardware, and validated end-to-end. It reconstructs a projectile's 3-D trajectory from three time-multiplexed range sensors and predicts its landing point in polar coordinates with a per-throw uncertainty. Over 31 ground-truthed throws the raw landing error is 93 mm on average; the trajectory carries a statistically significant 68 mm along-track bias that an empirical, significance-gated, throw-aligned drag correction removes, cutting the held-out mean error from 90 mm to 56 mm. The project also demonstrates sound measurement practice under an unexpected result: when the intended tennis-ball target proved acoustically undetectable, the failure was surfaced, escalated and resolved by an approved target substitution rather than silently worked around — and because every geometric correction is parametric in the ball radius, the change required no algorithmic modification. The reconstruction and prediction accuracy achieved with roughly €15 of sensors, dominated by a well-understood and partly-correctable drag systematic, meets the objectives set for the system.